

Models and Methods of Data Mining in Project Management

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Abstract — This article discusses Data Mining in the field of Project Management in terms of identifying hidden patterns or relationships between objects in large data sets. Theoretical foundations and tasks that are solved during the application of data mining are considered. The properties and features of existing methods of data mining are analyzed and described - from the construction of decision trees to neural networks. Tasks and problems in the field of project management are formulated, which can be solved using the methods of data mining.

Keywords — Data Mining, regression, classification, clustering, decision trees, neural networks, Project Management

I. INTRODUCTION

The rapid development and implementation of information technologies and software tools at the end of the 20th - beginning of the 21st century radically changed the approaches to project management (PM) in various fields, including the field of computer-aided design (CAD). The development of databases (DB) and database management systems (DBMS) has made it possible to store, process and present, at the request of the user, huge amounts of data related to projects conducted in various design organizations and products developed within the framework of these projects. The development of various methodologies and tools for project management (for example, Gantt charts) allowed both managers and participants to visualize and control the entire design process, and in case of deviations, make decisions on adjusting plans.

However, it should be noted that in connection with the accumulation of huge amounts of data in organizations, the question arises: how to effectively use this data in current and future projects in such a way as to speed up the development and maintenance of the latter? How to make sure that the entire amount of accumulated data does not lie "dead weight" in the organization's databases or archives - but "works"? The answer to this question can be various models and methods of data mining (DM), which are now increasingly being implemented in project management systems (PMS) and decision support systems (DSS).

This article is devoted to an overview of existing models and methods of DM, their properties and differences. The possibilities and prospects of using DM in the field of PM are considered, as well as the types of knowledge that can be obtained using data mining are identified. A comparison of DM methods for analyzing large amounts of data has been made and the most appropriate methods have been identified that would allow the most complete and accurate analysis of the data stored in design organizations.

II. DEFINITION OF DATA MINING

Data Mining fully reveals the specific and unique interdependencies of different data from each other. It allows you to quickly make the most correct decisions based on the available data. The second most important task facing data analysis is the discovery of new implicit patterns that are formed in data sets.

The most active intellectual analysis began to be used at the end of the 20th century. This was facilitated by rapid technological development. Information and data have begun to be treated as the most important and valuable resource that brings profit. Due to the significant development of technology, the amount of data in digital format has increased to incredible proportions. Intellectual analysis allows you to quickly and efficiently detect information of interest, quickly operate it to perform certain tasks.

At the moment, most scientists consider it a classic definition of data mining given by Grigory Pyatetsky-Shapiro: "Data mining is the study and detection of a "machine" (algorithms, artificial intelligence) in raw data of hidden knowledge that was not previously known, non-trivial, practically useful, accessible for interpretation" [1].

The following properties of knowledge follow from the considered definition:

1) **Knowledge must be new, previously unknown.** As you know, all the forces and resources that are used to obtain information that is already available never pay off. This confirms the fact that the most valuable resource in this area can only be new knowledge, which no one could get to before.

2) **Knowledge must be non-trivial.** Naturally, the data obtained must be non-trivial, unexpected. This is important when studying all the relationships between the elements and objects of the studied data. Only in case of detection of all hidden, previously unknown data, we can talk about some value of the IAD implementation. In other words, after receiving the results and establishing the fact of obtaining hidden data, all invested resources, including time, pay off in full. Data Mining is a very complex and time-consuming method of obtaining information and is not used in all cases. If there is a potential opportunity to obtain similar information in less costly ways, then data mining is not used.

3) **Knowledge should be practically useful.** After the discovery of new knowledge, it is very important to ensure that all this knowledge and information is somehow applicable in practice, bring appropriate benefits. In other words, Data Mining is used only when the potential data discovered after using the method will be useful for a particular person or organization.

4) **Knowledge must be accessible to human understanding.** It is important that all the information received be logically explained in simple language for a person. The value of new data lies in the fact that, despite its relative simplicity, the data obtained with the help of intellectual analysis does not cause unnecessary questions, contradictions or misunderstandings.

III. MODELS AND TASKS OF DATA MINING

A. Data Mining Models

Data Mining is a complex method for obtaining valuable new information. Naturally, for functioning in all areas, at all levels, it is important to properly prepare the key components of the method. In other words, to adapt all the resources of the method so as to get the maximum result. As a result, various forms of Data Mining are formed, which can be used depending on the conditions for the implementation of intellectual analysis.

The most common data mining models are:

- association rules;
- decision trees;
- clusters;
- mathematical functions.

Methods for forming Data Mining models are usually referred to the area of the so-called. "artificial intelligence".

B. Data Mining Tasks

Various Data Mining methods are used to solve problems such as [2]:

- descriptive;
- predictive.

Descriptive tasks focus entirely on deforming and changing all the data being analyzed for a more comfortable understanding. The most important point in such models is the maximum simplification of data and their transparency for understanding by an ordinary person. Also in this situation, it is worth noting separately that all descriptive tasks must necessarily simplify the information, even if the results obtained will be analyzed for the first time. The most common forms of descriptive tasks are association rules and clustering.

All *predictive* tasks are broken down into two fundamental steps by default. The first stage involves the formation of the necessary model of intelligent data, based on the nature and set of information that will be disclosed in this task. It is important to achieve the state of maximum productivity of the model.

If we focus on the ways to solve the tasks, we can distinguish two key ways:

- *supervised learning*;
- *unsupervised learning*.

In the first case, the implementation of the entire data analysis process will be divided into several corresponding stages. To begin with, a special Data Mining model is created, which is best revealed in the process of data analysis.

Such a model is called a classifier. Only after the creation of an appropriate classifier will its further study be of great importance. If an unsatisfactory result is obtained, an additional procedure for studying the classifier takes place. This type of problem is most often presented in the form of regression and classification.

The second case involves the process of holistically bringing together all the tasks that make it possible to define descriptive models. This type of problem is represented by clustering and searching for associative conditions (rules).

Let us study and analyze each of the mentioned problems of DM [3].

The task of **classification (stratification)** involves the study of all the features and characteristics of an object (event) in order to assign an appropriate class to it. Prior to the class determination process, all classes must be fully researched and limited in number. Only in this case can we talk about the subjectivity of the chosen class for the object (phenomenon).

The task of **regression** is a similar process to classification, except that all identification processes carried out are concentrated around one distinguishing parameter. In other words, one characteristic parameter allows you to determine the classification to which an object (phenomenon) can be attributed.

Both of the above forms (regression and classification) use two variables to establish a third, which will also represent a unique characterizing form of this very object [4].

If we consider regression or classification through the solution of the problem, then it is presented in this format. There is a certain set of different objects:

$$I = \{i_1, i_2, \dots, i_j, \dots, i_n\},$$

where i_j – object under study.

At the same time, each object also has its own specific variables that make it up:

$$I_j = \{x_1, x_2, \dots, x_h, \dots, x_m, y\},$$

where x_h - independent variables whose values are known and on the basis of which the value of the dependent variable is determined y .

Quite often, for a more comfortable representation of the form of dependent variables, a vector view is used:

$$X = \{x_1, x_2, \dots, x_h, \dots, x_m\}.$$

Each variable x_h can take values from some set:

$$C_h = \{c_{h1}, c_{h2}, \dots\}.$$

If the values of a variable are elements of a finite set, then it is said to have a categorical type. If the set of values $C = \{c_1, c_2, \dots, c_r, \dots, c_k\}$ of variable y is finite, then the problem is called the classification problem. If the variable y takes a value on the set of real numbers R , then the problem is called a regression problem.

The **clustering** problem is very similar to the previous example. Almost all the processes carried out directly in the formation of the classification problem are also used in clustering problems. However, in this case the set will be distributed into small subgroups (clusters) united by a

common parameter. It should be noted that during clustering, the final number of created clusters will be known only after the division of the set into groups is completed.

Formally, this problem is described as follows.

There is a set I with a certain amount of data. The problem lies in the need to form a set of clusters (subgroups) in accordance with certain parameters (attributes), so that there is a full mapping F of the set I to the set C , i.e. $F: I \rightarrow C$.

The set I is defined as follows:

$$I = \{i_1, i_2, \dots, i_j, \dots, i_n\},$$

where i_j - the object under study.

Each of the objects is characterized by a set of parameters:

$$i_j = \{x_1, x_2, \dots, x_h, \dots, x_m\}.$$

Each variable can take values from some set:

$$x = \{v_h^1, v_h^2, \dots\}.$$

The task of clustering is to construct a set:

$$C = \{c_1, c_2, \dots, c_k, \dots, c_g\}.$$

Here c_k - is a cluster containing objects similar to each other from the set I :

$$c_k = \{i_j, i_p \mid i_j \in I, i_p \in I \text{ и } d(i_j, i_p) < \sigma\},$$

where σ - a value that determines the measure of proximity to include objects in one cluster; $d(i_j, i_p)$ - a measure of proximity between objects, called *distance*.

A non-negative value $d(i_j, i_p)$ is called the *distance* between elements i_j and i_p , if the following conditions are met:

- a) $d(i_j, i_p) \geq 0$ for all i_j and i_p ;
- б) $d(i_j, i_p) = 0$ if and only if $i_j = i_p$;
- в) $d(i_j, i_p) = d(i_p, i_j)$;
- г) $d(i_j, i_p) \leq d(i_j, i_r) + d(i_r, i_p)$.

If the distance $d(i_j, i_p)$ is less than some value σ , then the elements are said to be close and placed in one cluster. Otherwise, the elements are said to be different from each other and they are placed in different clusters.

Most popular algorithms that solve the clustering problem use the difference matrix D as an input data format. The rows and columns of the matrix correspond to the elements of the set I . The matrix elements are the values $d(i_j, i_p)$ in row j and column p . Obviously, the values on the main diagonal will be zero:

$$D = \begin{pmatrix} 0 & d(e_1, e_2) & d(e_1, e_n) \\ d(e_2, e_1) & 0 & d(e_2, e_n) \\ d(e_n, e_1) & d(e_n, e_2) & 0 \end{pmatrix}.$$

Many studied and commonly used algorithms perform well with symmetric matrices. Provided that the absence of symmetry in the matrix is established, it is logical to carry out a typical transformation in this way:

$$(D + D^m) / 2$$

The task of **determining relationships (search for association rules)** is based on the process of separate selection of the most frequently encountered sets of objects among other objects of this very set. It is all the discovered relationships between objects that will be used to form the appropriate rules that greatly simplify the objects and phenomena being studied, make them as simple as possible to understand, and, accordingly, allow a person to predict further events associated with these components.

Naturally, the process of identifying the most common subsets of objects is very often used as a data mining problem.

In general, the task will look like this. To begin with, all objects related to the selected set of objects should be designated. This can be done in this form:

$$I = \{i_1, i_2, \dots, i_j, \dots, i_n\},$$

where i_j - objects included in the analyzed sets; n - total number of objects.

Sets of objects from the set I , stored in the database and subjected to analysis, are called *transactions*. Let us describe a transaction as a subset of the set I :

$$T = \{i_j \mid i_j \in I\}.$$

The set of transactions, information about which is available for analysis, is denoted by the following set:

$$D = \{T_1, T_2, \dots, T_r, \dots, T_m\},$$

where m - number of transactions available for analysis.

The set of transactions that include the object i_j , will be denoted as follows:

$$D_{ij} = \{T_r \mid i_j \in T_r; j = 1..n; r = 1..m\} \subseteq D.$$

Let us denote some arbitrary set of objects as follows:

$$F = \{i_j \mid i_j \in I; j = 1..n\}.$$

A set consisting of k objects is called a k -element set.

The set of transactions that include the set F is denoted as follows:

$$D_F = \{T_r \mid F \subseteq T_r; r = 1..m\} \subseteq D.$$

The ratio of the number of transactions that includes the set F to the total number of transactions is called the *support of the set F* and is denoted by $Supp(F)$:

$$Supp(F) = \frac{|D_F|}{|D|}.$$

The task of **sequence analysis (sequential analysis, search for sequential patterns)** is to find patterns among events interrelated in time.

Sequential analysis is quite interesting in terms of a more detailed and holistic study of all events, but in the correct chronological order. Such a procedure has a sufficient number of specific and unique processes. It is worth emphasizing the need for a clear identification of all patterns in the studied sequence. Such knowledge will contribute to the emergence of the ability to fully predict with a high degree of probability the events that will occur in the near future. It should be understood that such a process cannot be

perceived as an accurate and indisputable prediction of the behavior of objects or phenomena in the future, since it is practically impossible to foresee all external and internal circumstances that can affect this event or object, which already shifts the implementation of all processes far from the spectrum that provided for this analysis. In other words, such tasks show the most probable development of events, the logic of behavior that can happen in the absence of other independent circumstances that affect the course of development.

Based on all this, it is important to pay great attention to the concept of *sequence*. This term in the scientific literature explains any ordered set of objects. But in order for the set under study to acquire the characteristic features of a sequence, it is necessary to clearly establish any relation of order. In other words, when analyzing a set, one more property is simply added that affects this set, forcing all its elements to be examined in accordance with the order condition that was previously added.

Given all this information, you can show a set with a sequence condition in this form:

$$S = \{\dots, i_p, \dots, i_q, \dots\}, \text{ where } q < p.$$

It is very important to understand that the sequence is not a universal value, and for a more accurate characterization, the division into sequences with cycles and sequences without cycles should be taken into account. With cycles, a sequence is a sequence in which there is a repeating object at different positions:

$$S = \{\dots, i_p, \dots, i_q, \dots\}, \text{ where } q < p, i_q = i_p.$$

Sequence S support is the ratio of the number of transactions, which includes the sequence S , to the total number of transactions. A sequence is frequent if its support exceeds the minimum support set by the user:

$$Supp(S) > Supp_{min}.$$

The task of sequential analysis is to find all frequent sequences:

$$L = \{S \mid Supp(S) > Supp_{min}\}.$$

IV. APPLICATION OF DATA INTELLIGENCE METHODS IN PROJECT MANAGEMENT

The ultimate goal of extracting useful information in design activities is to increase the efficiency of engineers' work by reducing labor intensity and design time.

This is usually achieved by solving the following tasks of project activities:

1. Automation of paperwork.
2. Information support and automation of the decision-making process.
3. Unification of design decisions and design processes for reuse of design decisions, data and developments.
4. Strategic design.
5. Replacement of full-scale tests and prototyping by mathematical modeling.
6. Improving the quality of design management.

7. Application of alternative design and optimization methods.

Since the formulation of these tasks is as general as possible, the following is a brief description of more specific tasks in the field of planning and project management, where the use of IAD models and methods can be useful.

A. Project Resource Planning

Before starting a new project, its manager must determine some important parameters - the purpose of the project, the type of part or object being developed, the level of technical equipment, the qualifications of the performers, etc. Using models and methods of IAD, as well as the values of these parameters from other projects previously carried out by the organization, the project manager can obtain a preliminary estimate of the cost of resources for the implementation of the project - time, human, material, financial, etc.

All this data can be presented in the form of various graphs for analysis and informed decision making.

B. Quality control and project resources

In the course of the project, various situations may arise that in one way or another may affect the quality of the project or the cost of its resources. With the help of IAD models and methods, users can quickly receive analytical information about the progress of the project, the cost of its resources, etc., as well as control the occurrence of emergency situations. In addition, it is possible to obtain information about the links between events that led to certain consequences.

This information can be useful not only for analyzing the progress of the current project, but also for planning future projects of the organization.

C. Analysis of documents and project data

The application of data mining in this area can focus on analyzing the data warehouse of projects in order to find a similar configuration of project requirements that have already been completed by the organization. This can be achieved through text mining and rule generation using classification and data warehouse association projects [7]. IDA models and methods can be used to automatically summarize textual data and extract valuable rules that can be further transformed into a semantic network capable of providing a concise and accurate summary of the analyzed text.

D. Project Critical Path Analysis

Data mining also becomes extremely useful during the implementation phase of a project in situations where critical project activities are close to (or behind) schedule or when non-critical activities are approaching critical ones [8]. In this situation, data mining can be used to analyze similar actions from the data warehouse of current and previous projects and suggest alternative methods for performing a particular action to recover lost time at optimal cost.

The overall goal of applying data mining at this level is to ensure that the optimal economic solution for a project is realized when project circumstances change.

E. Project risk management

Data mining is extremely useful for facilitating project risk management. For example, risk identification is mainly concerned with the question: what can go wrong? The purpose of this process is to identify and specifically name the risks of the project and their characteristics [9].

Data mining can be applied by analyzing the project data warehouse, finding classification and association rules to determine the attributes of potential identified risks. The analysis should carefully examine the current project plan for areas of uncertainty compared to projects already implemented and look for problems that could cause the project to fall behind schedule.

The result of the analysis will be a complete list of risks, in which the risk can be classified according to a number of main features, consisting of two components, namely: the probable cause of a particular condition (for example, non-compliance with the delivery schedule by the subcontractor) and the overall impact of the risk on the project (for example, control points will not be achieved). points and/or the budget will be exceeded) [10].

F. Application of data mining methods to solve problems in the field of project management

Below is the connection of the above tasks, solved in the field of project management, with the main classes of tasks and methods of data mining.

1. Study of the design situation.

DM methods: *k-nearest neighbor method, Bayesian networks, nearest neighbor method.*

2. Search for analogues

DM methods: *k-nearest neighbors, Bayesian networks, nearest neighbors, associative memory.*

3. Search for inconsistencies and novelty

DM methods: *k-nearest neighbor method, associative memory.*

4. Link modeling

DM methods: *Bayesian networks, nearest neighbors method, forecasting.*

5. Search for connections

DM methods: *k-nearest neighbor method, associative memory, prediction.*

6. Evaluation of options for decision making

DM methods: *k-nearest neighbors, Bayesian networks, nearest neighbors, aggregation.*

7. Product quality analysis

DM methods: *decision trees, linear regression, k-nearest neighbors.*

8. Estimation of the cost of a new object on the market

DM methods: *linear regression, neural networks, genetic algorithms.*

9. Selection of performers

DM Methods: *Decision Trees*

10. Document analysis

DM methods: *Bayesian networks, nearest neighbors, neural networks, genetic algorithms.*

11. Planning the timing of the project

DM methods: *decision trees, neural networks.*

12. Estimate the timing of a specific task

DM methods: *k-nearest neighbors method.*

13. Estimating resource costs

DM methods: *linear regression, neural networks, k-nearest neighbors method.*

V. CONCLUSION

In this work, the definition of data mining was given, its main stages were highlighted. Various IAD methods are analyzed, their classification is given.

The study revealed that various IAD methods can be applied in the project activities of organizations in order to increase its speed and quality, as well as for more efficient project management and rational use of resources (material, financial, time, etc.). For each of the tasks faced by project managers and executors, the main IAD methods have been identified that can be successfully used to solve them.

REFERENCES

- [1] Telnov Yu.F. Intelligent information systems in the economy. M.: SINTEG, 2002. 306 p.
- [2] Barseghyan A. A., Kupriyanov M. S., Kholod I. I., Tess M. D., Elizarov S. I. Analysis of data and processes: textbook. allowance. St. Petersburg: BHV-Peterburg, 2009. 512 p.
- [3] Duke V.A., Samoylenko A.P. Data Mining: training course. St. Petersburg: Piter, 2001. 368 p..
- [4] Chubukova I. A. Data Mining: textbook. allowance. M.: Internet University of Information Technologies: BINOM: Knowledge Laboratory, 2006. 382 p.
- [5] Palmov S. V. Data mining: textbook. allowance. Samara: PSUTI, 2017. 124 p.
- [6] Stepanov R.G. Data Mining Technology: Data Mining. Kazan State University IN AND. Ulyanov-Lenina, 2008. 57 p.
- [7] Nemati, H. R., & Barko, C. D. (2002). Enhancing Enterprise Decisions Through Organizational Data Mining. Journal of Computer Information Systems, (Summer), 21-28.
- [8] Iranmanesh, S. H., & Mokhtari, Z. (2008, August). Application of Data Mining Tools to Predicate Completion Time of a Project. In Proceedings of World Academy of Science, Engineering and Technology, (Vol. 32, pp. 234-239).
- [9] Datta, R. P. (2008). Data Mining Applications and Infrastructural Issues: An Indian Perspective. ICAFI Journal of Infrastructure, 6(3), 42-50.
- [10] Turner, K., & Zizzamia, F. (2008). Predicting Better Claims Management. Risk Management, 55 (7), 52-55.
- [11] A. Bakir, B. Turhan, A. Bener, A Comparative Study for Estimating Software Development Effort Intervals, "Software Quality Journal" 2011, vol. 19, pp. 537-552.
- [12] P. Sentas, L. Angelis, I. Stamelos, Multinomial Logistic Regression Applied on Software Productivity Prediction, 9th Panhellenic Conference in Informatics, Thessaloniki 2003.
- [13] I. Barcelos Tronto, J. Simoes da Silva, N. Sant'Anna, Comparison of Artificial Neural Network and Regression Models in Software Effort Estimation, International Joint Conference on Neural Networks, Orlando 2007, pp. 771-776.